

Few-step Generative Models (Flow-Map models, ODE integrators)

Syllabus

Part I. Motivation. Why Few-Step Models?

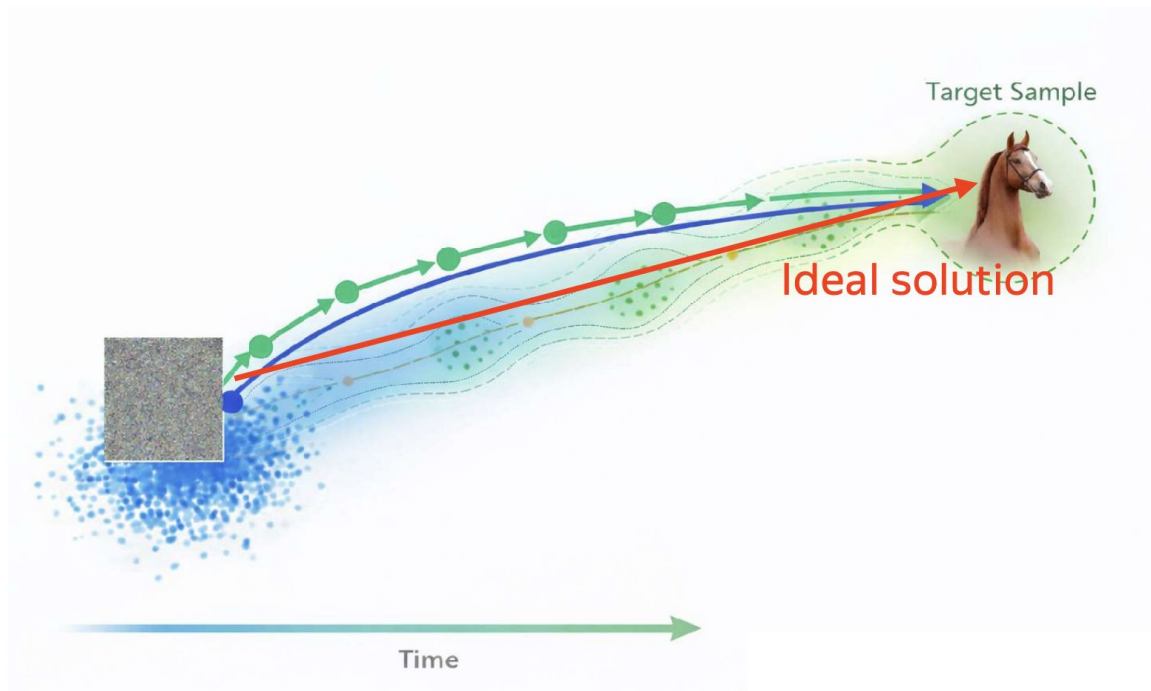
- Why DMs are slow?
- Naive solution: step knowledge distillation

Part II. Traditional view on few-step models

- Consistency Models (CMs)
- Multistep CMs (Multi-boundary)
- ShortCut Models
- Mean Flow Models

Part I. Motivation

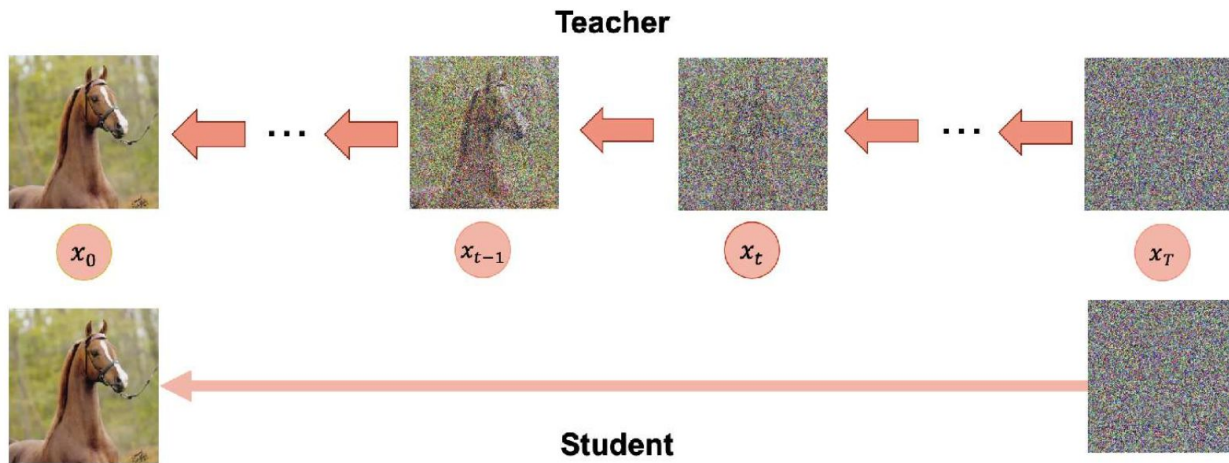
DMs have curved trajectories



Knowledge Distillation

Main idea.

- collect the data using diffusion $\{(x_\sigma^1, x_0^1), \dots, (x_\sigma^N, x_0^N)\}$
- fine-tune the network $x_\theta(x)$ on this data, $x_\theta(x_\sigma) \rightarrow x_0$



Knowledge Distillation

$$L_{diff}(\mathbf{x}_\phi) = \mathbb{E}_{t \sim \mathcal{U}(0, \sigma)} \mathbb{E}_{p_{data}(\mathbf{x}_0)} \mathbb{E}_{\mathcal{N}(\mathbf{x}|\mathbf{x}_0, t^2)} \|\mathbf{x}_\phi(\mathbf{x}, t) - \mathbf{x}_0\|_2^2$$

usual diffusion

$$L_{KD}(\mathbf{x}_\theta) = \mathbb{E}_{\mathcal{N}(\mathbf{x}|0, \sigma^2)} \|\mathbf{x}_\theta(\mathbf{x}) - \hat{\mathbf{x}}_\phi(\mathbf{x})\|_2^2$$

$\hat{\mathbf{x}}_\phi(\mathbf{x})$ – sample from the pretrained DM using solver

one-step KD (perfect),
holy grail of speed

$$L_{KD_m}(\mathbf{x}_\theta) = \mathbb{E}_{t \sim \mathcal{U}(0, \sigma)} \mathbb{E}_{p_{data}(\mathbf{x}_0)} \mathbb{E}_{\mathcal{N}(\mathbf{x}|\mathbf{x}_0, t^2)} \|\mathbf{x}_\theta(\mathbf{x}, t) - \hat{\mathbf{x}}_\phi(\mathbf{x}, t)\|_2^2$$

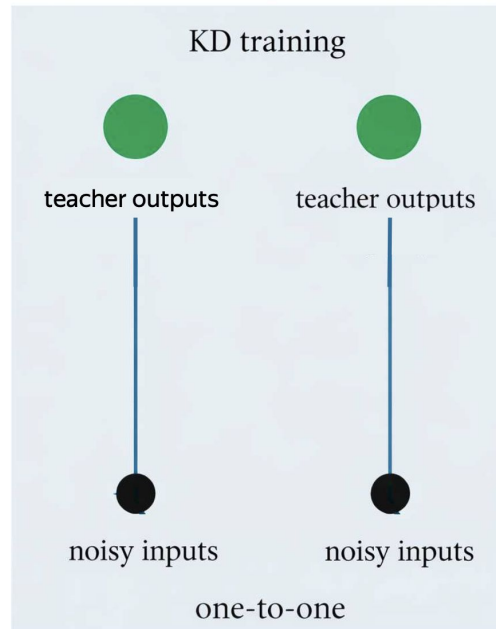
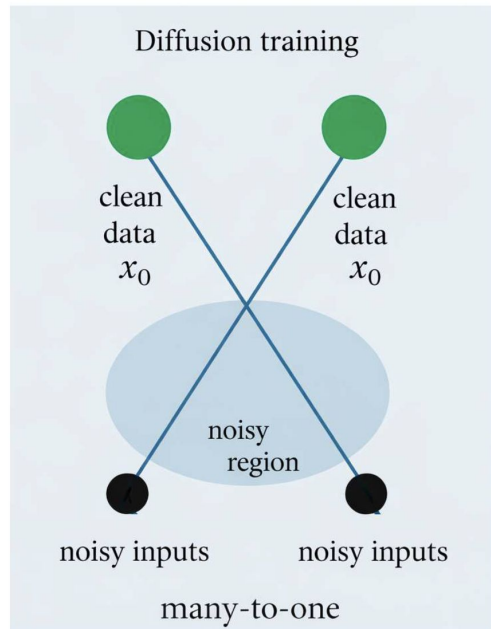
multi-step KD (progressive
distillation, student is
trained to mimic teacher's
output after 2 steps of ODE
solver in 1 step)

Why is the Diffusion slow but KD fast? Intuition

Diffusion:

Forward process is one-to-many
(x_0 can evolve into an infinite number of noisy versions x_t , depending on specific random noise sampled)

Reverse process is many-to-one
(it is the challenge model faces during training L_{diff})



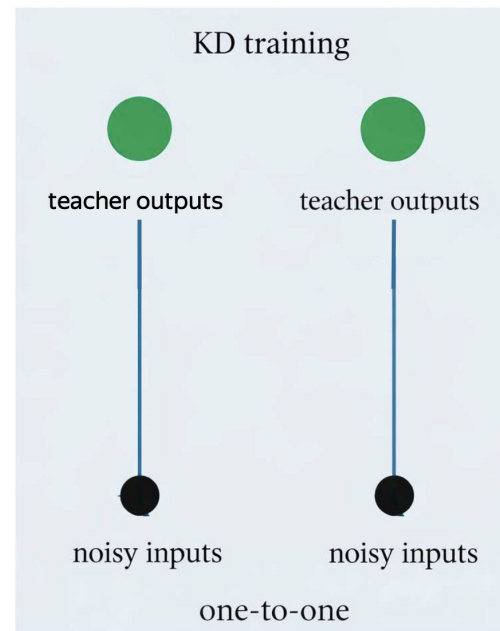
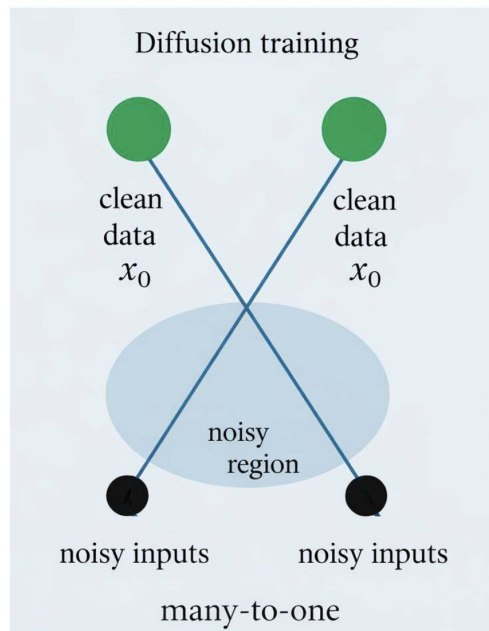
Why is the Diffusion slow but KD fast? Intuition

Diffusion:

Because of many-to-one problem, we can't solve diffusion in 1 step.

At high noise levels, the model can't know which specific x_0 to pick, so it just predicts a very vague "global average."

As we take small steps and t decreases, the noise becomes less dominant. The model slowly commits to one path



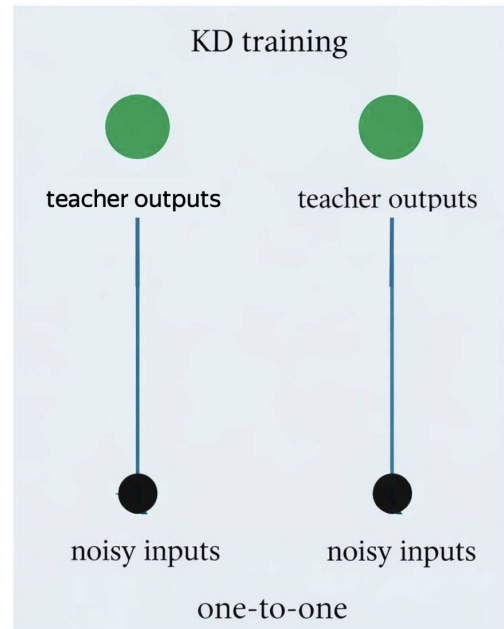
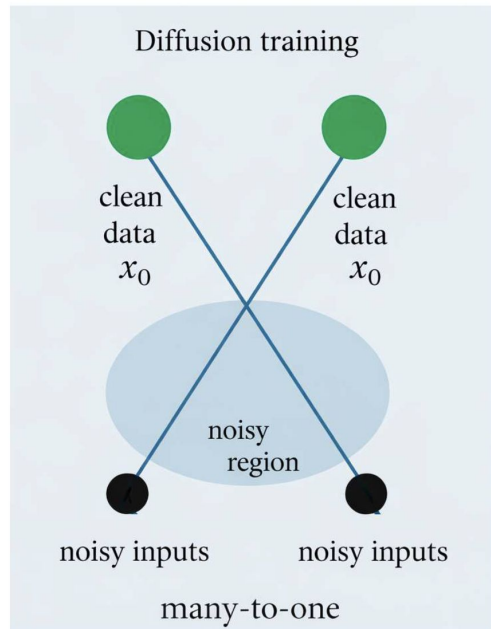
Why is the Diffusion slow but KD fast? Intuition

Knowledge Distillation:

Turns this into a **One-to-One** mapping.

Since the teacher's ODE is **deterministic**, each noisy input maps to exactly one teacher output.

The teacher model has already "resolved" the ambiguity of the data distribution. The objective is much easier for a neural network to learn in a single jump without blurriness.



So, we need to learn a direct mapping along the ODE trajectory instead of its velocity...

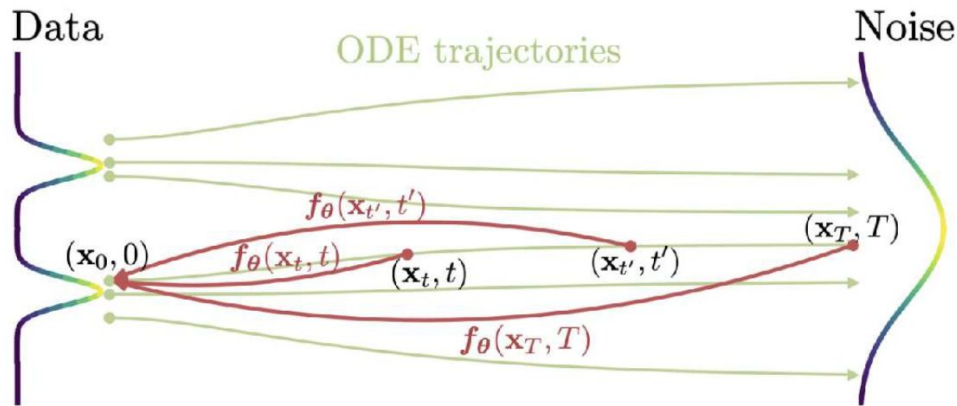
Part II. Consistency Models

What is a mapping?

Definition. Consistency function

Given a solution trajectory of the PF ODE, we define the **consistency function** as

$$f : (x_t, t) \rightarrow x_0, \forall x_t \in \{x_\tau\}_{\tau \in [0, \sigma]},$$



How to learn Consistency Mapping?

Self-consistency property.

$$f_{\theta}(x_t, t) = x_0 \quad \forall x_t \rightarrow f_{\theta}(x_0, 0) = \dots = f_{\theta}(x_s, s) = \dots = f_{\theta}(x_{\sigma}, \sigma)$$

No matter where you are on a specific solution path (at time t, s , or σ), the model must map you to the **exact same starting point x_0** .

This ensures that the model isn't just "denoising" (moving slightly closer to the data), but "projecting" (jumping all the way to the origin of that specific path)

How to learn Consistency Mapping?

Consider discretizing the time horizon into N sub-intervals

The **objective** for learning the Consistency Mapping:

$$\mathbb{E} \left\| f_{\theta}(\mathbf{x}_{n-1}, t_{n-1}) - f_{\theta}(\mathbf{x}_n, t_n) \right\|_2 \rightarrow \min_{\theta}$$

Any issues?

How to learn Consistency Mapping?

Consider discretizing the time horizon into N sub-intervals

The **objective** for learning the Consistency Mapping:

$$\mathbb{E} \left\| f_{\theta}(\mathbf{x}_{n-1}, t_{n-1}) - f_{\theta}(\mathbf{x}_n, t_n) \right\|_2 \rightarrow \min_{\theta}$$

Any issues?

1. We might have a degenerate solution. For example, $f_{\theta}(\mathbf{x}_t, t) = 0$
2. How to get $\{\mathbf{x}_n\}$ from the ODE solution?

How to learn Consistency Mapping?

Consider discretizing the time horizon into N sub-intervals

The **objective** for learning the Consistency Mapping:

$$\mathbb{E} \left\| f_{\theta}(\mathbf{x}_{n-1}, t_{n-1}) - f_{\theta}(\mathbf{x}_n, t_n) \right\|_2 \rightarrow \min_{\theta}$$

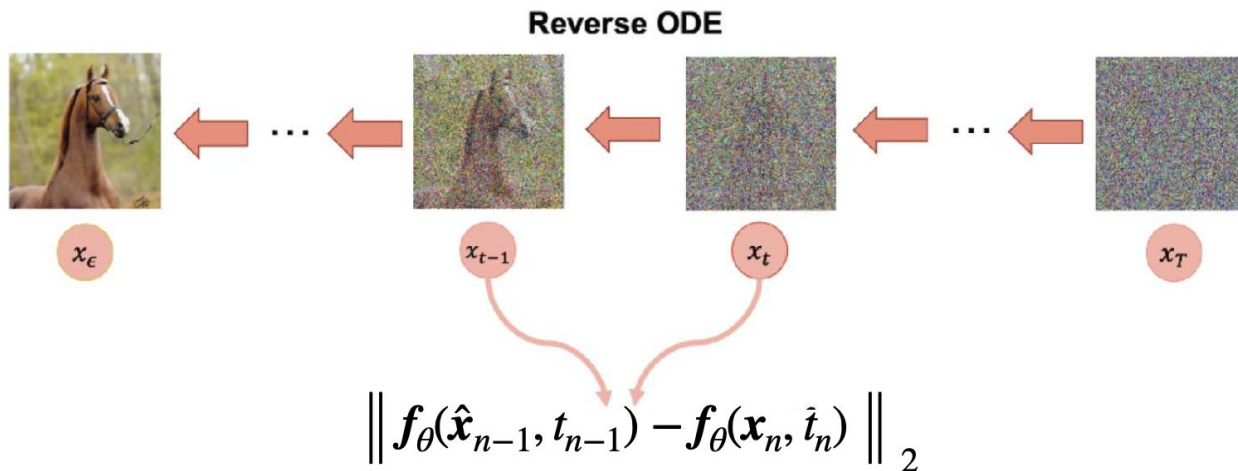
How to fix:

1. Define a **boundary condition** $f_{\theta}(\mathbf{x}_0, t_0) = \mathbf{x}_0$
2. Use a teacher model: $\mathbf{x}_n \sim \mathcal{N}(\mathbf{x}_n | 0, t_n^2)$; $\hat{\mathbf{x}}_{n-1} = \Phi(\mathbf{x}_n, t_n, t_{n-1} | \phi)$

Consistency Distillation. Summary

$$\mathcal{L}_{CD} = \mathbb{E}_{t_n \sim \mathcal{U}(t_0, \dots, t_N)} \mathbb{E}_{\mathbf{x}_0 \sim p_{data}(\mathbf{x}_0)} \mathbb{E}_{\mathbf{x}_n \sim \mathcal{N}(\mathbf{x}_n | \mathbf{x}_0, t_n^2)} \left\| f_{\theta}(\hat{\mathbf{x}}_{n-1}, t_{n-1}) - f_{\theta}(\mathbf{x}_n, t_n) \right\|_2$$

$$\hat{\mathbf{x}}_{n-1} = \Phi(\mathbf{x}_n, t_n, t_{n-1} | \phi), f_{\theta}(\mathbf{x}_0, t_0) = \mathbf{x}_0$$



Consistency Training (different from CD)

$$\left\| f_{\theta}(\mathbf{x}_{n-1}, t_{n-1}) - f_{\theta}(\mathbf{x}_n, t_n) \right\|_2$$

What if we no longer have a teacher?

We still do $\mathbf{x}_n \sim \mathcal{N}(\mathbf{x}_n | 0, t_n^2)$, but how to get $\mathbf{x}_{n-1}$ now?

$$d\mathbf{x} = -t \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) dt$$

$$\nabla_{\mathbf{x}} \log p_t(\mathbf{x}) = -\frac{1}{t^2} \left(\mathbf{x} - \mathbb{E}[\mathbf{x}_0 | \mathbf{x}] \right)$$

Can we estimate this?

Consistency Training (different from CD)

$$d\mathbf{x} = -t \nabla_x \log p_t(\mathbf{x}) dt$$

$$\nabla_x \log p_t(\mathbf{x}) = -\frac{1}{t^2} \left(\mathbf{x} - \mathbb{E} [\mathbf{x}_0 | \mathbf{x}] \right) \approx -\frac{1}{t^2} (\mathbf{x} - \mathbf{x}_0)$$

$$d\mathbf{x} = \frac{1}{t} (\mathbf{x} - \mathbf{x}_0) dt$$

$$\int_{t_n}^{t_{n-1}} \frac{d(\mathbf{x} - \mathbf{x}_0)}{(\mathbf{x} - \mathbf{x}_0)} = \frac{dt}{t}$$

$$\mathbf{x}_{n-1} = \frac{t_n}{n} \mathbf{x}_{n-1} - \mathbf{x}_0 \left(\frac{t_n - t_{n-1}}{t_{n-1}} \right) = \mathbf{x}_0 + t_n \epsilon$$

Consistency Training (different from CD)

$$\mathcal{L}_{CT} = \mathbb{E}_{t_n \sim \mathcal{U}(t_0, \dots, t_N)} \mathbb{E}_{\mathbf{x}_0 \sim p_{data}(\mathbf{x}_0)} \mathbb{E}_{\mathbf{x}_n \sim \mathcal{N}(\mathbf{x}_n | \mathbf{x}_0, t_n^2)} \left\| \mathbf{f}_{\theta}(\mathbf{x}_{n-1}, t_{n-1}) - \mathbf{f}_{\theta}(\mathbf{x}_n, t_n) \right\|_2$$

$$\mathbf{x}_n = \mathbf{x}_0 + t_n \epsilon \quad \mathbf{x}_{n-1} = \mathbf{x}_0 + t_{n-1} \epsilon$$

We simply sample points from the linear path, where $\epsilon \sim \mathcal{N}(0, I)$ is **the same noise vector**. This ensures they belong to the same "straightened" trajectory.

Consistency Training (CT) "forces" the trajectories to be straight

Sampling from Consistency Models

Can we apply ODE solvers to CMs?

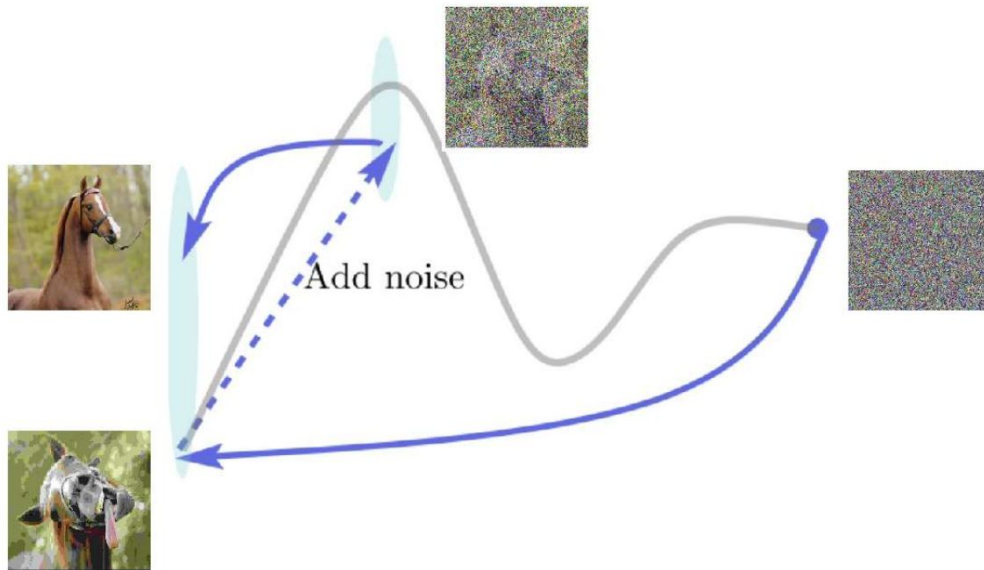
$f_{\theta}(x_n, t_n)$ is **not** a score function anymore - **we cannot use solvers to generate data** :(

CM is **not a vector field**; it is a **flow map** that has already "pre-integrated" the ODE. It doesn't tell you where to move next; it tells you where you will eventually land.

While a score function is a **derivative** (which an ODE solver is designed to integrate), a consistency model is the **solution to that integral**.

Sampling from Consistency Models

Solution: **stochastic multistep sampling!**



$$\mathbf{x}_0 = f_{\theta}(\mathbf{x}_N, \sigma), \mathbf{x}_N \sim \mathcal{N}(0, \sigma^2)$$

for $n = 1$ **to** $N - 1$ **do**

 Sample $\epsilon \sim \mathcal{N}(0, 1)$

$$\mathbf{x}_n = \mathbf{x}_0 + t_n^2 \epsilon$$

$$\mathbf{x}_0 = f_{\theta}(\mathbf{x}_n, t_n)$$

We need about 4-5 steps

Multi-boundary (Multistep) CMs

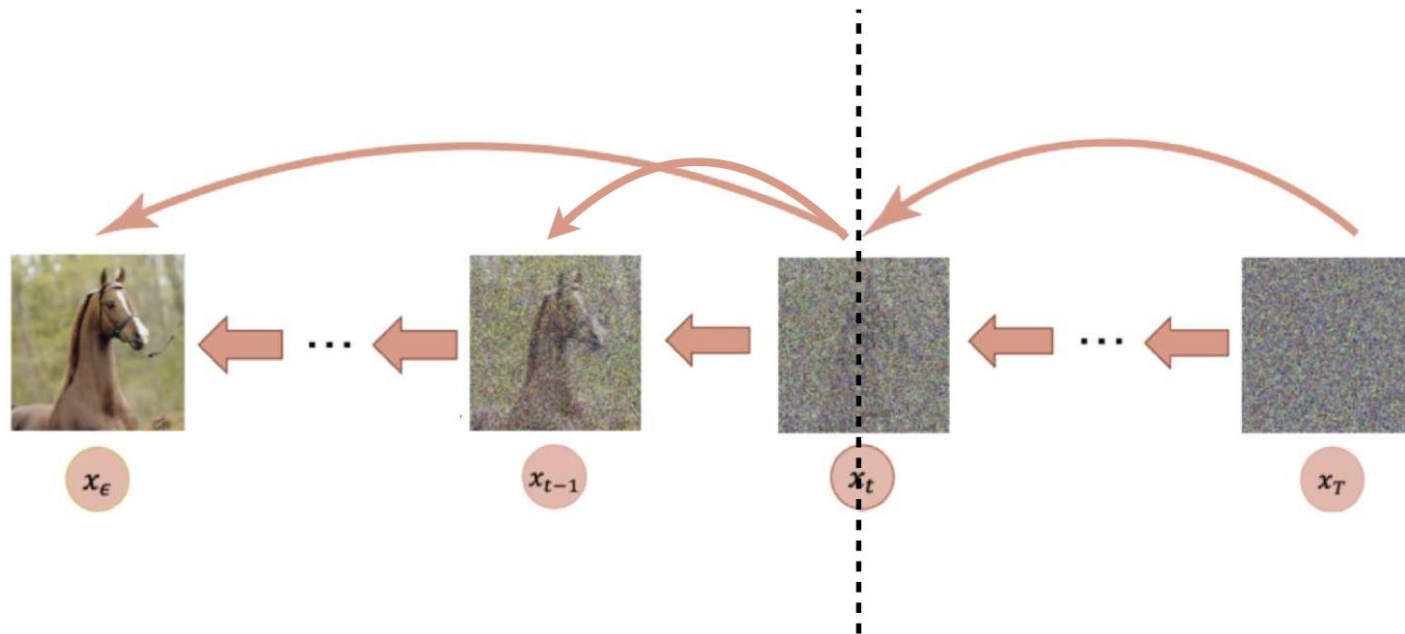
The core issue is that we need to approximate the entire solution interval

Can we solve that?

Given a solution trajectory of the PF ODE on the interval $[t, s]$ we define the **multistep consistency function** as

$$f : (x_r, r, s) \rightarrow x_s, \forall x_r \in \{x_\tau\}_{\tau \in [t, s]}$$

Multi-boundary (Multistep) CMs



How to learn Multistep CMs?

1. Split the time horizon $[0, \sigma]$ into N intervals
N-parts of self-consistency properties:

$$[t_0, t_s] : f_{\theta}(\mathbf{x}_t, t, t_s) = \mathbf{x}_s \quad \forall \mathbf{x}_t \rightarrow f_{\theta}(\mathbf{x}_0, t_0, t_s) = \dots = f_{\theta}(\mathbf{x}_s, t_s, t_s)$$

...

$$[t_m, \sigma] : f_{\theta}(\mathbf{x}_t, t, \sigma) = \mathbf{x}_m \quad \forall \mathbf{x}_t \rightarrow f_{\theta}(\mathbf{x}_m, t_m, \sigma) = \dots = f_{\theta}(\mathbf{x}_{\sigma}, \sigma, \sigma)$$

2. Learn a CM on each interval

$$\mathbb{E}_{t_n \sim \mathcal{U}(t_0, \dots, t_N)} \mathbb{E}_{\mathbf{x}_0 \sim p_{data}(\mathbf{x}_0)} \mathbb{E}_{\mathbf{x}_n \sim \mathcal{N}(\mathbf{x}_n | \mathbf{x}_0, t_n^2)} \left\| f_{\theta}(\mathbf{x}_{n-1}, t_{n-1}, t'_n) - f_{\theta}(\mathbf{x}_n, t_n, t'_n) \right\|_2$$

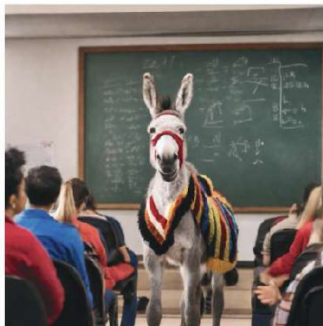
How to learn Multistep CMs?

Ours, 4 steps



*A donkey in a clown costume giving a lecture at the front of a lecture hall.
The blackboard has mathematical equations on it.*

Teacher, 50 steps

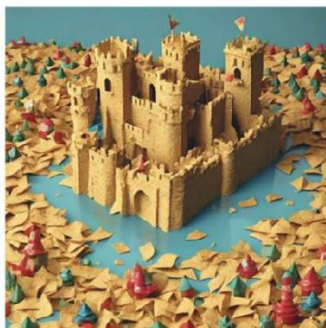


Ours, 4 steps

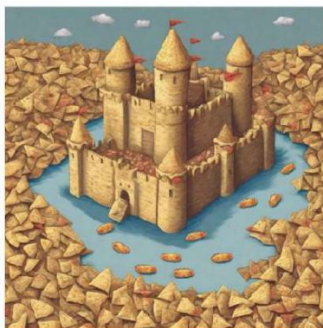


an airplane taxiing on a runway with the sun behind it

Teacher, 50 steps



*A castle made of tortilla chips, in a river made of salsa.
There are tiny burritos walking around the castle*



*A raccoon wearing formal clothes, wearing a top hat and holding a cane.
The raccoon is holding a garbage bag. Oil painting in the style of cubism.*



MeanFlows, ShortCut...

After consistency models, a huge amount of work came out

Stable consistency tuning: Understanding and improving consistency models

[FY Wang, Z Geng](#), H Li - arXiv preprint arXiv:2410.18958, 2024 - arxiv.org

Diffusion models achieve superior generation quality but suffer from slow generation speed due to the iterative nature of denoising. In contrast, consistency models, a new generative ...

☆ Save ⓘ Cite Cited by 10 Related articles All 4 versions ⓘ

One-step diffusion distillation via deep equilibrium models

[Z Geng, A Pople, JZ Koller](#) - Advances in Neural ..., 2023 - proceedings.neurips.cc

Diffusion models excel at producing high-quality samples but naively require hundreds of iterations, prompting multiple attempts to distill the generation process into a faster network ...

☆ Save ⓘ Cite Cited by 68 Related articles All 7 versions ⓘ

One-step Diffusion Models with f -Divergence Distribution Matching

[Y Xu, W Nie, A Vahdat](#) - arXiv preprint arXiv:2502.15681, 2025 - arxiv.org

Sampling from diffusion models involves a slow iterative process that hinders their practical deployment, especially for interactive applications. To accelerate generation speed, recent ...

☆ Save ⓘ Cite Cited by 19 Related articles All 2 versions ⓘ

Align your flow: Scaling continuous-time flow map distillation

[A Sabour, S Fidler, K Kreis](#) - arXiv preprint arXiv:2506.14603, 2025 - arxiv.org

Diffusion-and flow-based models have emerged as state-of-the-art generative modeling approaches, but they require many sampling steps. Consistency models can distill these ...

☆ Save ⓘ Cite Cited by 31 Related articles All 3 versions ⓘ

Consistency models made easy

[Z Geng, A Pople, W Luo, J Lin, JZ Koller](#) - arXiv preprint arXiv:2406.14548, 2024 - arxiv.org

Consistency models (CMs) offer faster sampling than traditional diffusion models, but their training is resource-intensive. For example, as of 2024, training a state-of-the-art CM on ...

☆ Save ⓘ Cite Cited by 133 Related articles All 4 versions ⓘ

One step diffusion via shortcut models

[K Frans, D Hafner, S Levine, P Abbeel](#) - arXiv preprint arXiv:2410.12557, 2024 - arxiv.org

Diffusion models and flow-matching models have enabled generating diverse and realistic images by learning to transfer noise to data. However, sampling from these models involves ...

☆ Save ⓘ Cite Cited by 222 Related articles All 3 versions ⓘ

Consistency flow matching: Defining straight flows with velocity consistency

[L Yang, Z Zhang, Z Zhang, X Liu, M Xu...](#) - arXiv preprint arXiv ..., 2024 - arxiv.org

Flow matching (FM) is a general framework for defining probability paths via Ordinary Differential Equations (ODEs) to transform between noise and data samples. Recent ...

☆ Save ⓘ Cite Cited by 74 Related articles All 3 versions ⓘ

Mean flows for one-step generative modeling

[Z Geng, M Deng, X Bai, JZ Koller, K He](#) - arXiv preprint arXiv:2505.13447, 2025 - arxiv.org

We propose a principled and effective framework for one-step generative modeling. We introduce the notion of average velocity to characterize flow fields, in contrast to ...

☆ Save ⓘ Cite Cited by 231 Related articles All 3 versions ⓘ

Inductive moment matching

[L Zhou, S Ermon, J Song](#) - arXiv preprint arXiv:2503.07565, 2025 - arxiv.org

Diffusion models and Flow Matching generate high-quality samples but are slow at inference, and distilling them into few-step models often leads to instability and extensive ...

☆ Save ⓘ Cite Cited by 77 Related articles All 5 versions ⓘ

Consistent diffusion models: Mitigating sampling drift by learning to be consistent

[G Daras, Y Dagan, A Dimakis...](#) - Advances in Neural ..., 2023 - proceedings.neurips.cc

Imperfect score-matching leads to a shift between the training and the sampling distribution of diffusion models. Due to the recursive nature of the generation process, errors in previous ...

☆ Save ⓘ Cite Cited by 75 Related articles All 7 versions ⓘ

Multistep consistency models

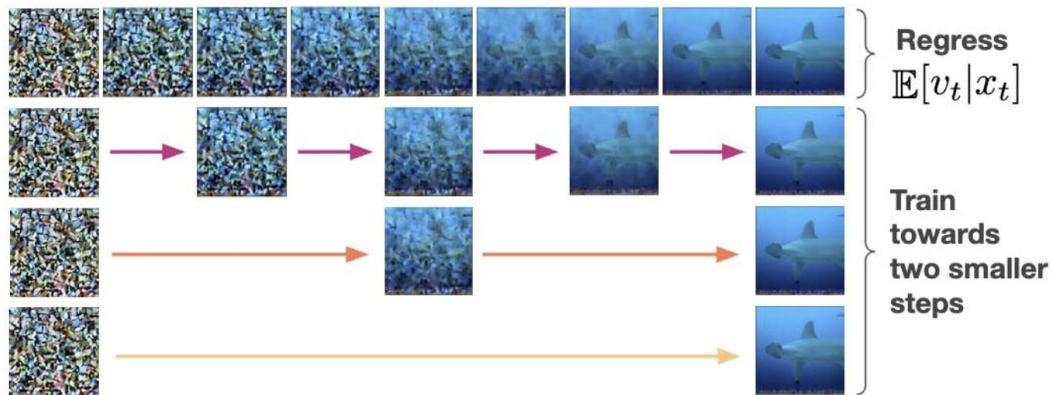
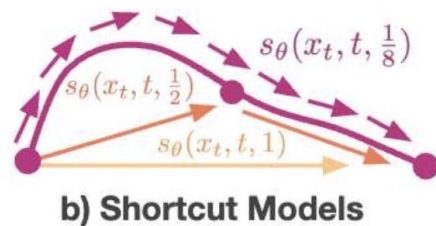
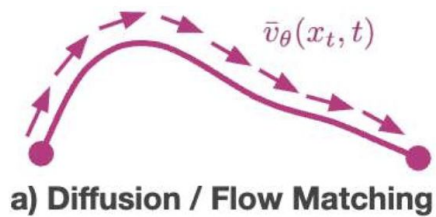
[J Heek, E Hoogeboom, T Salimans](#) - arXiv preprint arXiv:2403.06807, 2024 - arxiv.org

Diffusion models are relatively easy to train but require many steps to generate samples.

Consistency models are far more difficult to train, but generate samples in a single step. In ...

☆ Save ⓘ Cite Cited by 74 Related articles All 3 versions ⓘ

ShortCut models



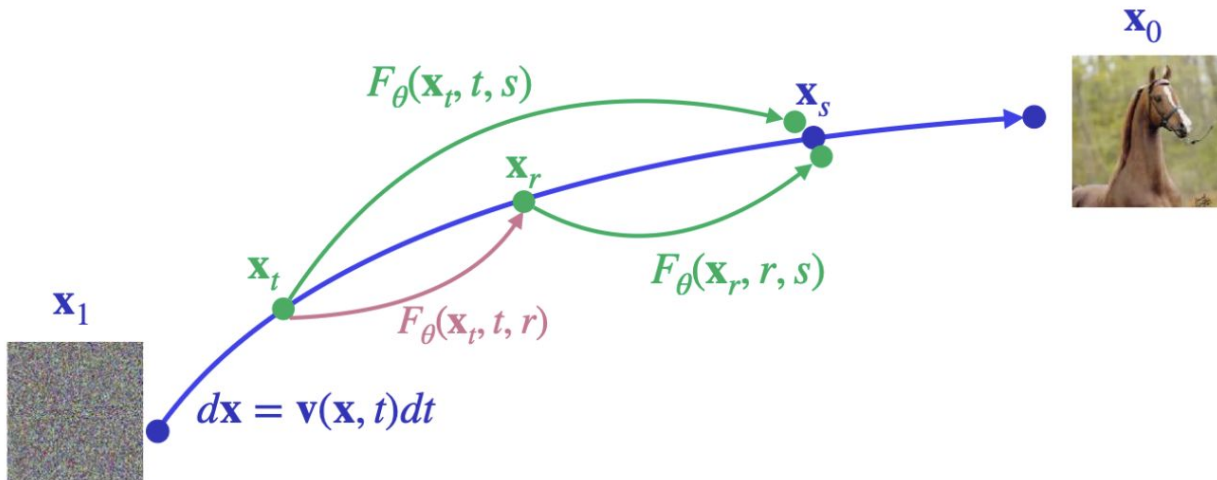
ShortCut models

Interval additivity property

$$\int_t^s \mathbf{v}(\mathbf{x}, u) \, du = \int_t^r \mathbf{v}(\mathbf{x}, u) \, du + \int_r^s \mathbf{v}(\mathbf{x}, u) \, du$$

Objective

$$\|F_\theta(\mathbf{x}_t, t, s) - sg(F_\theta(F_\theta(\mathbf{x}_t, t, r), r, s))\|_2^2$$



Mean Flow (2025)

$$F_{\theta}(\mathbf{x}_t, t, s) \approx \frac{1}{s-t} \int_t^s \mathbf{v}(\mathbf{x}, u) du \Rightarrow \Psi_{t \rightarrow s}(\mathbf{x}_t) \approx \mathbf{x}_t + (s-t)F_{\theta}(\mathbf{x}_t, t, s)$$

In standard Flow Matching or Diffusion, the model learns the **instantaneous velocity**

The **Mean Flow** model is trained to approximate the average value of that velocity over the interval $[t, s]$

If you know the average velocity over the whole interval, you can calculate the destination $\Psi_{t \rightarrow s}(\mathbf{x}_t)$ directly

Mean Flow (2025)

$$(s - t)F_{\theta}(\mathbf{x}_t, t, s) = \int_t^s \mathbf{v}(\mathbf{x}, u) du$$



Mean Flow Identity

$$F_{\theta}(\mathbf{x}_t, t, s) = \underbrace{\mathbf{v}(\mathbf{x}_t, t)}_{\text{Inst. velocity}} - (s - t) \underbrace{\frac{d}{dt} F_{\theta}(\mathbf{x}_t, t, s)}_{\text{Time derivative}}$$

Avg. velocity

Goal: train $F_{\theta}(\mathbf{x}_t, t, s) \approx \frac{1}{s - t} \int_t^s \mathbf{v}(\mathbf{x}, u) du$, objective:

$$\|F_{\theta}(\mathbf{x}_t, t, s) - sg\left(\underbrace{\mathbf{v}(\mathbf{x}_t, t) - (s - t) \frac{d}{dt} F_{\theta}(\mathbf{x}_t, t, s)}_{\text{Mean Flow Identity}}\right)\|_2^2 + L_{FM}$$

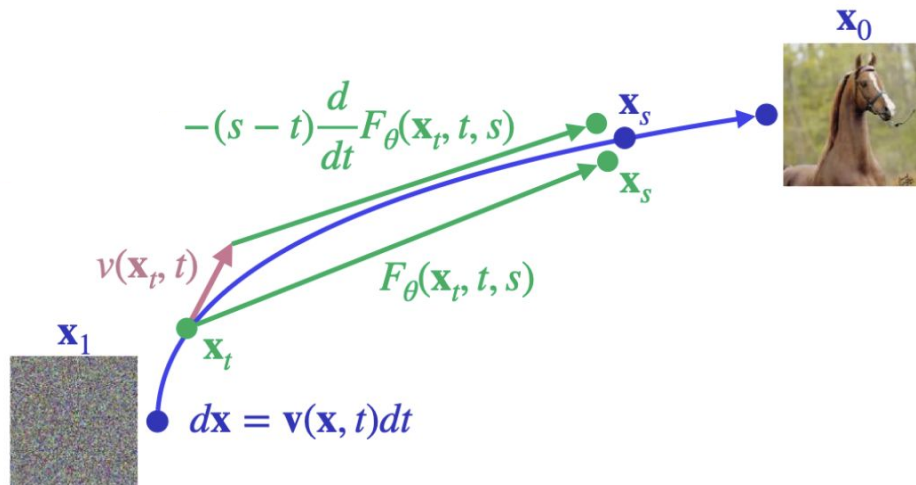
$$F_{\theta}(\mathbf{x}_t, t, t) = \mathbf{v}_{\theta}(\mathbf{x}_t, t)$$

Mean Flow (2025)

$$\|F_{\theta}(\mathbf{x}_t, t, s) - sg\left(\mathbf{v}(\mathbf{x}_t, t) - (s - t)\frac{d}{dt}F_{\theta}(\mathbf{x}_t, t, s)\right)\|_2^2 + L_{FM}$$

Mean Flow Identity

$$F_{\theta}(\mathbf{x}_t, t, t) = \mathbf{v}_{\theta}(\mathbf{x}_t, t)$$



АЙТИШНИКИ



ВСЁ!